App Dev Project Report

1. Student Details

Name: Aaditya Chaturvedy
Roll Number: 24F2008937
Email: 24F2008937@ds.study.iitm.ac.in
About Me: I am a student at IIT Madras BS Degree program and SRM Institute of Technology with interest in edge AI and embedded deep learning. My work focuses on deploying production-grade ML models on resource-constrained hardware such as ARM devices.

2. AI/LLM Declaration

I used **Gemini (Google)** strictly for syntax debugging, generating boilerplate code and UI/CSS alignment. All core architectures and feature integrations were developed manually. The extent of AI/LLM usage is around **10-15%**. All final correction and implementation logic were done manually.

3. Description

Project Title: IIT Placement Portal Application

Developed a centralized Flask-based web application to streamline campus recruitment for institutes, companies, and students. Utilized SQLite for relational data management and Jinja2 with Bootstrap to build dynamic, role-specific dashboards. Automated the recruitment lifecycle by managing company approvals, tracking student applications, and maintaining historical placement records.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Core backend web framework
Flask-SQLAlchemy	Object Relational Mapper for database interaction
Wekzeug	Security utilities, including secure password hashing
Jinja2	Template engine for rendering dynamic HTML pages
HTML5 & CSS	Core markup structure and custom frontend styling
Bootstrap 5	Frontend styling framework for responsive design
Chart.js	Data visualization for admin reports and trends
SQLite	Lightweight local database for storing application data

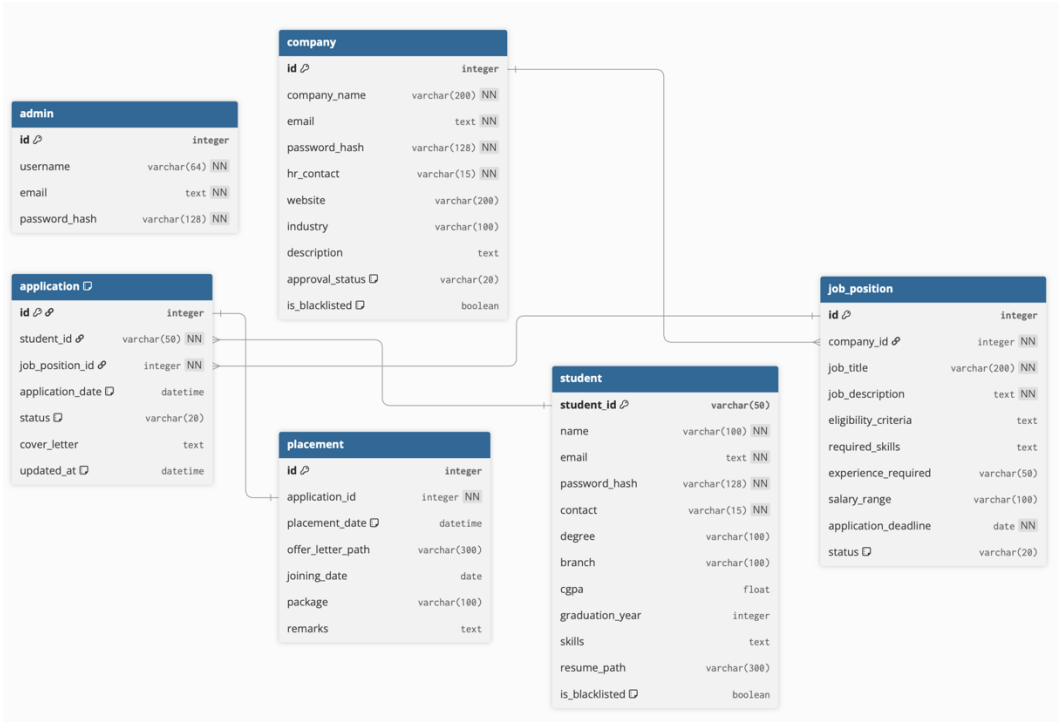
5. DB Schema Design

Tables:

1. **Admin:** stores superuser credentials for the institute placement cell (id, username, email, password_hash)
2. **Company:** stores recruiter organization details and their system access status (id, company_name, email, password_hash, hr_contact, website, industry, description, approval_status, is_blacklisted)
3. **Student:** stores student profiles, academic records, and resume links (student_id, name, email, password_hash, contact, degree, branch, cgpa, graduation_year, skills, resume_path, is_blacklisted)
4. **JobPosition:** stores details of placement drives created by approved companies (id, company_id, job_title, job_description, eligibility_criteria, required_skills, experience_required, salary_range, application_deadline, status)
5. **Application:** stores records of student submissions to specific placement drives (id, student_id, job_position_id, application_date, status, cover_letter, updated_at)
6. **Placement:** stores final hiring records and salary packages for selected students (id, application_id, placement_date, offer_letter_path, joining_date, package, remarks)

Relationships:

- One-to-Many → ***Company*** → ***JobPosition***
- One-to-Many → ***Student*** → ***Application***
- One-to-Many → ***JobPosition*** → ***Application***
- One-to-One → ***Application*** → ***Placement***



6. API Resource Endpoints

Endpoint	Method	Description
/api/companies	GET	Fetches a list of companies
/api/students	GET	Fetches a list of students
/api/jobs	GET	Fetches a list of available jobs
/api/jobs	POST	Creates a new job listing
/api/applications	GET	Fetches a list of applications
/api/jobs/<job_id>/apply	POST	Submits a job application for a specific job ID
/api/applications/<app_id>/status	PATCH	Updates the status of a specific job application ID

YAML API Definition File:

Included separately in the submission ZIP as **api.yaml** .

7. Architecture and Features

Architecture Overview:

- **app.py** – Main Flask application entry point, containing controllers and routing logic
- **models.py** – Database models and schema definitions using SQLAlchemy
- **helpers.py** – Utility functions, decorators, and validation logic
- **/templates** – Jinja2 HTML templates, organized into subdirectories by user role (`admin`, `company`, `student`)
- **/static** – CSS styling, JavaScript files, and static assets used across the platform

Implemented Features:

- Role-based user registration and login (Admin, Company, Student)
- Secure student profile management with resume document uploads
- Job creation, editing, and timeline management for recruiting companies
- Real-time job application submission and applicant status tracking (Applied, Shortlisted, Selected, etc.)
- Dedicated analytics and data visualization dashboards for companies and administrators
- Secure password hashing and robust user session handling

Additional Features:

- Export placement records and system data (e.g., CSV exports for admin reports)
- System moderation capabilities for admins (Company approval workflows, student/company blacklisting)
- Internal RESTful API endpoints to fetch resource data dynamically

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1wMn5UGP6-UI4o3E6m-nummeq5zS1bCMk/view?usp=share_link